

The Agentic Harness



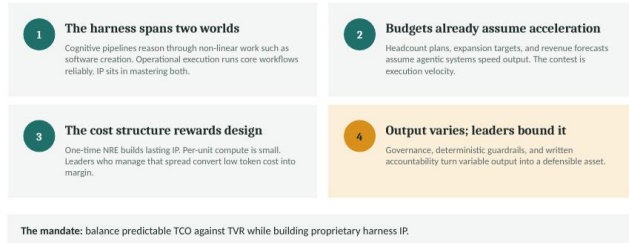
Everyone here has used a chatbot. Today is about something different: the agentic harness. Models are rented and commoditizing. The durable IP is the harness: the architecture, rules, and controls that turn organizational know-how into repeatable, defensible execution. The harness governs two worlds: cognitive pipelines that reason through non-linear work such as software creation, and operational execution that runs core workflows with deterministic reliability. As a CTO I build from the ground up, so today we focus on operational execution and total cost of ownership. By the end you will draw a harness on paper and defend where your IP resides.

Direct the leverage; own the agentic IP

THE ASK | THE EXECUTIVE MANDATE

Direct the leverage; own the agentic IP

We need leaders who direct enterprise scale and build agentic IP.



The Agentic Harness | 2

Architecture and leadership move balance sheets; enthusiasm does not. Four facts define the mandate. First, the harness blends two worlds: cognitive pipelines that reason through open-ended work such as software engineering, and operational execution that runs routine workflows reliably. Second, your growth targets already assume agentic acceleration; the contest is execution velocity and market capture. Third, the cost structure rewards design: a one-time design spend yields a low marginal cost of execution, and the spread between token spend and value created becomes margin. Fourth, models are probabilistic. Your job is to bound that variance, control it, and direct it toward economic yield. We need business architects who balance predictable TCO against TVR while building proprietary harness IP.

Three decisions define the harness

EXECUTIVE SUMMARY

Three decisions define the harness

1

Own the harness; rent the model

Models commoditize. Your IP is the architecture, rules, and controls that make output repeatable and defensible.

2

Code by default; AI for judgment

If two top performers always give the same answer, write it in code. Pay for inference only where judgment creates value.

3

Budget the full lifecycle

Tokens are one line of seven. Review, platform, and assurance drive TCO. Measure cost per completed decision.

Session objective: draw an autonomous harness on paper and defend where your IP resides.

The Agentic Harness | 3

Three decisions carry the session. First, the model is rented; the harness is owned. Second, code is the default and AI is reserved for judgment. Third, the business case covers the full lifecycle, and the unit of measure is cost per completed decision. Everything after this slide supports one of these three.

Value and cost are separate levers

THE ECONOMICS

Value and cost are separate levers

The slide is divided into two main sections. The left section, titled 'THE CEILING' in an orange header, is for 'Total Value Realized' and lists benefits like velocity, throughput, speed, and margin. The right section, titled 'TODAY'S FOCUS' in a teal header, is for 'Total Cost of Ownership' and lists operational costs like OpEx, infrastructure spend, and reliability. A footer note states that the session covers TCO and TVR modeling.

THE CEILING	TODAY'S FOCUS
Total Value Realized <i>Top-line and margin upside</i> • Velocity: cycle times from weeks to minutes • Throughput: higher volume with existing teams • Speed: proposals and products iterated in real time • Margin: spread between compute spend and yield	Total Cost of Ownership <i>The operating floor</i> • OpEx, token consumption, and system overhead • Predictable, optimized infrastructure spend • Grounded baseline and cost containment • Reliability that makes the TVR target credible

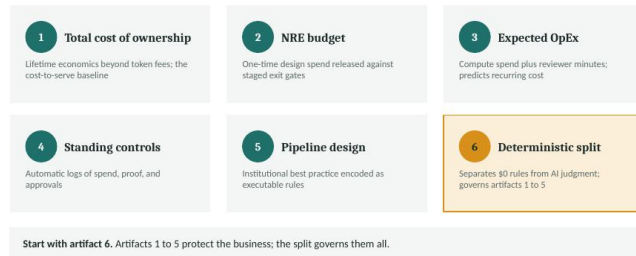
Advantage is the spread. This session covers TCO; TVR modeling gets its own session.

Total Value Realized is the ceiling: revenue expansion, velocity, and margin unlocked when intelligent systems execute at scale. Total Cost of Ownership is the floor: what it takes to run those systems reliably. Advantage is the spread between the two. We set the TVR target here, then spend the rest of the session on TCO. TVR modeling gets a dedicated session.

Six artifacts form the operating blueprint

THE BLUEPRINT

Six artifacts form the operating blueprint



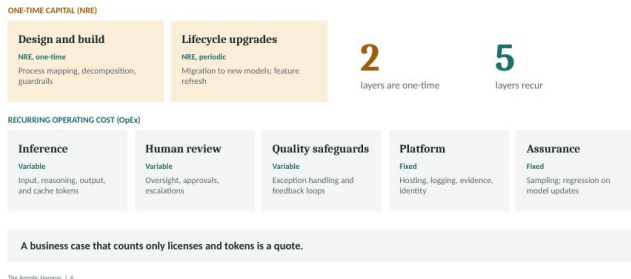
The Agentic Harness | 5

These six artifacts are the components of the live system you will see. Artifacts one through five are the ledgers, budgets, controls, and pipelines that protect the business. Artifact six, the deterministic split, governs the other five, so that is where the hands-on work starts. The next slides take them in order of importance, not number.

Budget all seven cost layers, not just tokens

ARTIFACT 1 | TOTAL COST OF OWNERSHIP

Budget all seven cost layers, not just tokens



The full TCO stack has seven layers. Token charges are typically the smallest line; human oversight and platform assurance form the operational core. Two layers are one-time capital: design and build, and periodic lifecycle upgrades. Five recur. Treat a vendor model release as a scheduled re-verification cost, not an emergency. A business case that counts only licenses and tokens is a quote, not a TCO.

Deterministic by default; probabilistic where it pays

ARTIFACT 6 | THE DETERMINISTIC SPLIT

Deterministic by default; probabilistic where it pays

Deterministic		Probabilistic	
<i>The floor</i>		<i>The multiplier</i>	
BEHAVIOR	Same input, same output, every time	BEHAVIOR	Synthesis; output varies by run
MARGINAL COST	\$0 per run once coded	MARGINAL COST	Token charge per run
ASSURANCE	Fully testable and auditable	ASSURANCE	Requires guardrails and review
USE FOR	Code, SQL, thresholds, checksums, routing, arithmetic	USE FOR	Unstructured text, market signals, exceptions, drafting

Executive test: if two top performers always give the same answer, write the rule in code. AI output never executes unchecked; code validates it before capital moves.

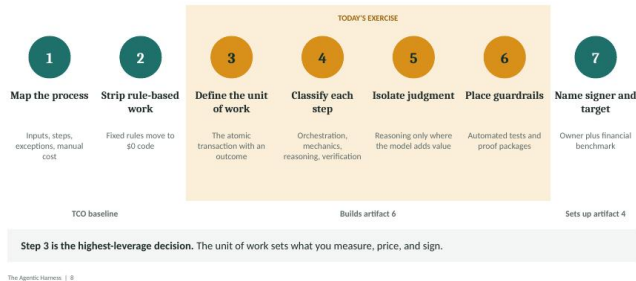
The Agentic Harness | 7

This is the most important slide. It is a capital allocation decision. Deterministic steps cost nothing per run once coded, behave identically every time, and satisfy auditors. Probabilistic steps provide reasoning but charge per execution and vary. Default to code for routine logic and deploy AI where judgment creates value. If two competent managers always handle a step the same way, it is a business rule, not a prompt. Moving rules from prompts into code lowers the operating floor permanently.

Seven steps convert a process into a harness

THE METHOD

Seven steps convert a process into a harness

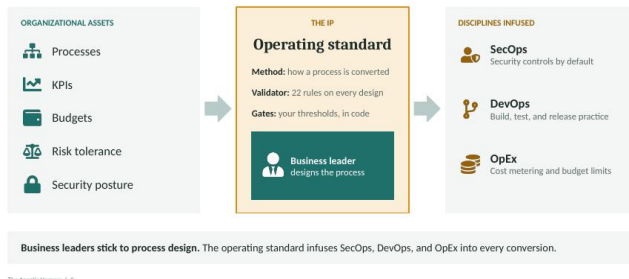


This seven-step method converts a business process into an agentic system. Steps one and two establish the TCO baseline. Step three, defining the unit of work, is the highest-leverage decision. Step five isolates judgment so you pay for tokens only where reasoning creates value. Step seven names the signer and the financial target. In the workshop you apply steps three through six to a live process on paper.

The operating standard is the IP

THE OPERATING STANDARD | ORGANIZATIONAL IP

The operating standard is the IP



The demo you are about to see runs under one operating standard. It has three parts: a method that converts a described process into a design, a validator that checks every design against 22 rules, and gates that apply the company's thresholds in code. The standard carries the organization's assets: processes, KPIs, budgets, risk tolerance, and security posture. Because those assets live in the standard, the business leader sticks to their knitting: process design. SecOps, DevOps, and OpEx practice arrive with every conversion. The model is rented and available to every competitor. The operating standard is owned, and it is the IP.

Validators prove the standard is implemented

THE OPERATING STANDARD | ENFORCEMENT

Validators prove the standard is implemented

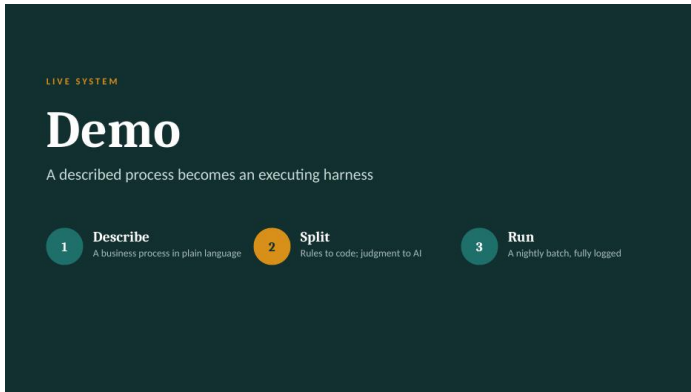
1 Screen the input Before any model reads it	2 Validate the design 22 rules; any error blocks	3 Gate every run Company thresholds, in code
• Too short or too long: refused, next action named • Text addressed to the system: refused • Demo: an invoice memo that told the system to skip a check was refused • Cost: zero tokens	V3 Named approver; the machine never approves its own work V12 Exactly one thinking step V13 Every judgment followed by a test or gate V15 Gate conditions state a number, a comparison, or "never" V16 Every refusal names a next human action V17 Code writes the proof; the model never does	• Variance $\leq \\$50$ and $\leq 2\%$ closes by arithmetic • Exposure above \$2,000 needs a second approver • No purchase order or supervised supplier: refused • Judgment arithmetic recomputed: demo caught -72.00 claimed against -720.00

Why validate: certainty that your specifics are implemented, every design and every run. Discussion: what are your approver, thresholds, and single decision point?

The Agentic Harness | 10

Why do we run these checks? To be certain the organization's specifics are implemented, on every design and every run, at zero token cost. Three layers. First, the input is screened by a rule before any model reads it; in the demo, an invoice memo that told the system to skip a check was refused. Second, the validator checks every generated design against 22 rules and blocks on any error; two you will recognize: V3 requires a named approver, because the machine never approves its own work, and V12 requires exactly one thinking step, one decision point per unit of work. Third, gates apply the company's thresholds in code during every run. Discovery questions for the room: Who is your named approver? What variance closes without judgment? What exposure needs a second approver? Which decision must stay at one? Each answer is a line in your operating standard.

Demo

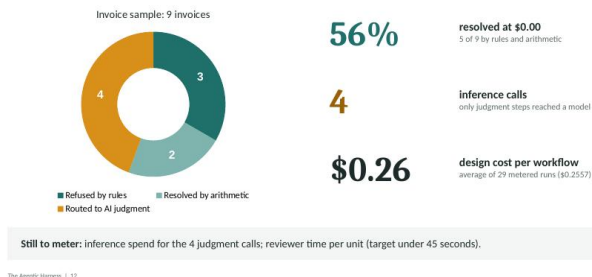


Switch to the live system. Three moves. First, describe a real business process in plain language. Second, watch the harness split it: fixed rules become code at zero marginal cost; judgment steps route to the model. Third, run a nightly batch and show the cost ledger and evidence store filling in. The next slide shows the numbers from that run.

Code resolved 5 of 9 invoices before any AI call

RESULTS | USE CASE: INVOICE SAMPLE

Code resolved 5 of 9 invoices before any AI call

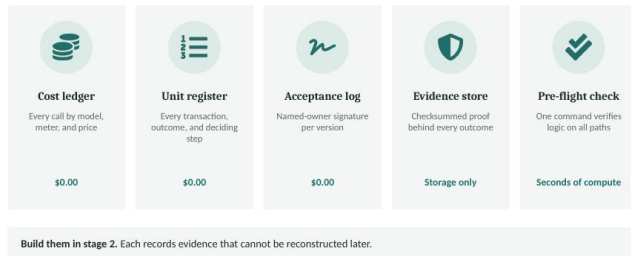


Use case: invoice sample. This is live data from the system behind this session. Of nine incoming invoices, deterministic rules refused three and arithmetic completed two, at zero token cost. Four required judgment and went to inference. Average design generation cost is 25.6 cents per workflow across 29 metered runs; 34 designs were generated in total, five of them stubs. Two numbers are not shown yet: inference spend for the four judgment calls, and reviewer time per unit. Those are metered in your own environment; an estimated figure would sit beside measured ones, so they remain open.

Five deterministic controls prove the result

ARTIFACT 4 | STANDING CONTROLS

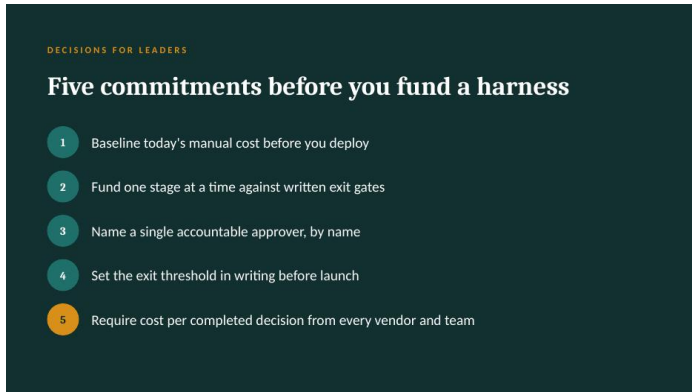
Five deterministic controls prove the result



The Agentic Harness | 13

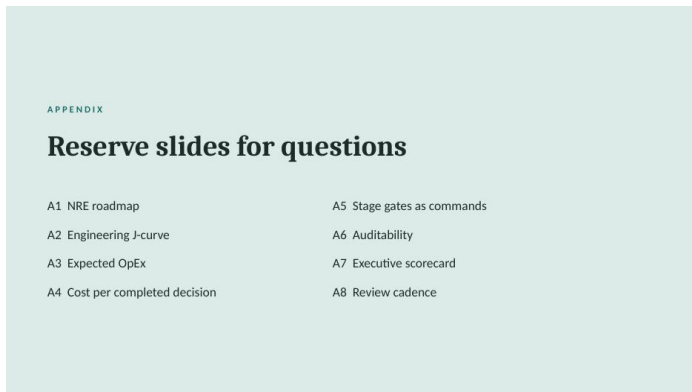
Artifact four replaces quarterly guesswork with continuous control. All five assets are deterministic and consume no tokens. They log exact spend, record proof of every decision, enforce named sign-off, and verify integrity before every release. Build them in stage two, during design; each records evidence that cannot be reconstructed later, and that evidence is what proves the result to boards and auditors.

Five commitments before you fund a harness



Close on commitments, not enthusiasm. Capture the manual baseline first, or the result cannot be proven. Release NRE one stage at a time. Name a single-threaded approver while enthusiasm is high. Write the exit threshold down before launch; that number is the clearest signal a proposal is honest. Shift procurement from billable hours and API calls to verified completed decisions. Reserve slides follow for questions.

Reserve slides for questions

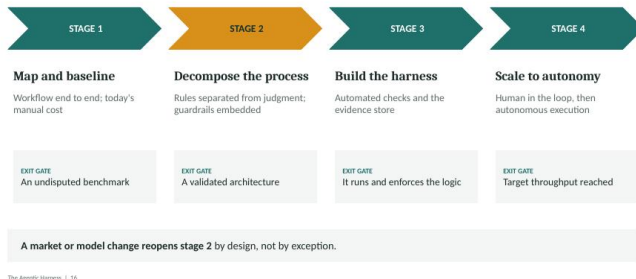


Do not present these in sequence. Each answers a question that reliably comes from an executive audience; the NRE roadmap and the J-curve are the most likely. If nobody asks, skip the appendix.

Fund the build one stage at a time

APPENDIX A1 | ARTIFACT 2

Fund the build one stage at a time



The Agentic Harness | 16

NRE is upfront design capital. Four stages, each with an exit condition. Stage one sets an undisputed baseline so every efficiency gain is measurable. Stage two is the architectural work that creates leverage. Stages three and four move from prototype to autonomous execution. A market or model change reopens stage two intentionally.

Protect the capacity that builds the harness

APPENDIX A2 | ARTIFACT 2 CONTINUED

Protect the capacity that builds the harness

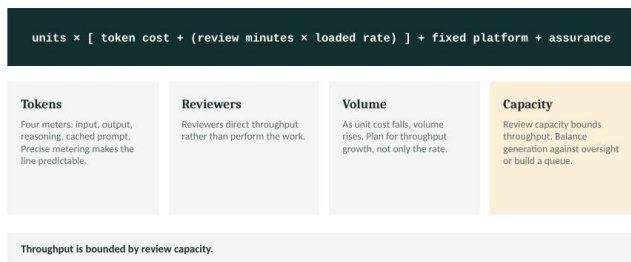


Harness engineering is unfamiliar work with little market experience behind it. Told only to do more with less, teams default to ad hoc scripts and hacked prompts, which is technical debt, not IP. Leaders need NRE air cover and should sell the J-curve: velocity dips while engineers learn harness patterns and build reusable standards, then the exchange rate improves. In this system, generating a validated workflow design averages about 26 cents per run. The curve shown is conceptual, not measured.

Expected OpEx: tokens and oversight as one equation

APPENDIX A3 | ARTIFACT 3

Expected OpEx: tokens and oversight as one equation



The Agentic Harness | 18

Operating cost combines compute and human expertise. Token costs are small and falling; human oversight protects quality. When a harness makes a task much cheaper, demand for that task rises: a top-line opportunity and a capacity problem at once. Plan review capacity so your best managers direct throughput instead of performing the work.

Measure cost per completed decision, not cost per request

APPENDIX A4 | UNIT ECONOMICS

Measure cost per completed decision, not cost per request

$$\text{token compute} + (\text{review minutes} \times \text{loaded rate}) + (\text{QA rate} \times \text{exception cost})$$

Term 1: architecture

300+ raw alerts into prompts nightly, versus routine data resolved in code and 12 decisions sent to AI.

Term 2: productivity

Structured evidence packages cut review from 20 minutes to 30 seconds.

Term 3: compounding

Verified, defect-free execution earns delegation to higher autonomy.

Examples on this slide are illustrative.

Diligence standard: every vendor and internal team reports cost per completed decision, not API calls.

The Agentic Harness | 19

Cost per completed decision is the unit metric. Term one is architecture: filter noise deterministically and reserve reasoning for what needs it. Term two is where productivity is won: structured evidence packages shorten review. Term three compounds: verified execution earns higher autonomy. The alert and review-time figures on this slide are illustrative examples, not measurements from this system.

The five stage gates became commands

APPENDIX A5 | FROM MEETINGS TO CONTROLS

The five stage gates became commands

Classic gate	Business question	Where it lives now	Executed by
Product requirements	Do we agree on goals and scope?	Process definition and unit-of-work map	Business leaders
Preliminary architecture	Is the design sound?	Workflow visualizer and step classifier	Leader reviewing an AI draft
Detailed design	Is the system complete?	22 structural checks, instantly	Deterministic engine
Readiness and proof	Are outputs provably accurate?	Quantitative gate conditions; auto-proof tests	Deterministic engine
Enterprise release	Fit to ship; who accepts it?	Cryptographic sign-off by a named executive	Named business leader

People own requirements and acceptance. Machines verify structure on every iteration.

The Agentic Harness | 20

Operational rigor stays; it becomes automated. Five design review gates once took months of calendar coordination and paper sign-off. Gates three and four now run in seconds as machine checks on every iteration, leaving immutable logs. Leaders stay in control where judgment matters: intent at the start and acceptance at release.

Automated design controls answer the auditor

APPENDIX A6 | AUDITABILITY

Automated design controls answer the auditor

Auditor's question	Harness answer	Evidence
What changed in the logic?	Version-controlled workflows; unapproved edits block execution	Diffable history
Who authorized it?	Gate requires a named owner's signature	Signed acceptance log
Was approval first?	Chained ledger: back-dating breaks verification	Tamper-evident sequence
Does execution match design?	Build verifies live controls against the spec	Automated parity proof
Were outputs uncorrupted?	Checksummed at generation; re-verified continuously	Cryptographic proof logs
Who can modify controls?	Separation of duties enforced in code	Role-based audit trail

SOX §404, 21 CFR Part 11, HIPAA §164.312. A vendor model update is a logic change: log the version and re-verify.

The Agentic Harness | 21

For public companies and regulated industries, the harness embeds verification in execution: every change, approval, and output leaves a tamper-evident trail. The new issue is the last line. A vendor model upgrade changes your control environment without notice. Treat it as a change event and re-verify automatically. Inference: few organizations do this today, and it is likely to become an audit finding.

The executive scorecard

APPENDIX A7 | ARTIFACT 4 CONTINUED

The executive scorecard

KPI	Source	Decision It drives
Cost per completed decision	Cost ledger + unit register	Expand, re-scope, or scale the process
Oversight minutes per unit	Unit register	Reallocate talent to higher-value work
Verified pass rate	Unit register + evidence store	Promote to a higher autonomy tier
Exception and refusal rate	Unit register	Tune rules and risk parameters
NRE burn against stage exits	Cost ledger	Approve stage funding or adjust build pace
Escaped defects (target: zero)	Evidence store	Go or no-go quality check

Baseline first. Without today's manual metrics, the result cannot be proven.

The Agentic Harness | 22

Every metric here originates in an automated control; a KPI with no asset behind it is an intention. Watch the refusal rate: a system that flags exceptions shows its guardrails work, and one that has never refused anything has never been tested. Capture the manual baseline before deployment, and align vendor contracts to verified completed outcomes.

Continuous review holds the result

APPENDIX A8 | REVIEW CADENCE

Continuous review holds the result

Frequency	Review focus	Owner
Every run	Exception flags, refusal logs, guardrail alerts	Process owner
Monthly	Cost per completed decision against target	Business leader
Quarterly	Oversight time; whether to raise the autonomy tier	Leader and process owner
On model change	Full verification suite re-run	Lead systems architect
On process change	Stage 2 reopens: new design, new signature	Business leader

Name the approver and write the exit threshold while enthusiasm is high.

Governance continues after go-live: monthly unit cost, quarterly oversight and autonomy review, and automatic re-verification whenever a vendor updates a model. Name a single-threaded owner and set the exit threshold before launch.